

Generalized linear models

GLMs for different distributions

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GLMs for different distributions - Contents

Continuous distributions

Discrete distributions

Grouped data

Continuous distributions

1. Normal distribution ✓

2. Exponential distribution:

- ▶ Suitable e.g. for $y \geq 0$, since the normal distribution does not guarantee to only predict positive values (also not for $\mu = \exp(\mathbf{x}^\top \beta)$) !
- ▶ Frequently used for modeling waiting times / duration times

Continuous distributions

$$Y \in \mathbb{R}_+$$

$$f(y) = \lambda e^{-\lambda y}$$

2. Exponential distribution (continued)

$$Y \sim \text{Exp}(\lambda), \quad \lambda \geq 0$$

$$f(y) = \lambda \exp(-\lambda y) = \exp(-\lambda y + \log(\lambda)) = \exp\left(\frac{y(-\lambda) - (-\log(\lambda))}{1}\right)$$

$$\vartheta = -\lambda$$

$$b(\vartheta) = -\log(\lambda) = -\log(-\vartheta)$$

$$\phi = 1$$

$$c(y, \phi) = 0$$

Continuous distributions

2. Exponential distribution (continued)

$$\mathbb{E}(y) = b'(\eta) = \frac{-1}{-\eta} \cdot (-1) = \frac{1}{\eta} = \frac{1}{\lambda} \quad (\eta = \lambda)$$

$$\begin{aligned}\text{Var}(y) &= \phi b''(\eta) = 1 \cdot \left(-\frac{\partial}{\partial \eta} \frac{1}{\eta} \right) = - \left(-\frac{1}{\eta^2} \right) = \frac{1}{\eta^2} \\ &= \frac{1}{\lambda^2}\end{aligned}$$

→ Linkage between expectation and variance:

λ small $\leftrightarrow \mu = \frac{1}{\lambda}$ large $\leftrightarrow$ mean large
 $\leftrightarrow$ variance large

Continuous distributions

$$\mu_i = \frac{1}{\eta} \Leftrightarrow \eta_i = -\frac{1}{\mu_i}$$

2. Exponential distribution (continued)

What would be the canonical link?

$$\eta_i = -\frac{1}{\mu_i} = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad g(\mu_i) = -\frac{1}{\mu_i} = -\eta$$

$$\Rightarrow \mu_i = -\frac{1}{\mathbf{x}_i^\top \boldsymbol{\beta}} = -\frac{1}{\eta_i}$$

Problematic, since $\mu_i > 0$ has to hold.

Preferable in this case: (more stable also)

$$\mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$$

response fct.

$$\Leftrightarrow \log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

log = link

Discrete distributions

$$p(y) = \frac{\lambda^y}{y!} e^{-\lambda}$$

1. Bernoulli distribution ✓
2. Poisson distribution:

$$Y \sim \text{Poi}(\lambda), \quad \lambda > 0$$

$$f(y) = P(Y = y) = \frac{\lambda^y}{y!} \exp(-\lambda) = \exp(y \log(\lambda) - \log(y!) - \lambda)$$

$$\vartheta = \log(\lambda)$$

$$b(\vartheta) = \lambda = \exp(\vartheta)$$

$$\phi = 1$$

$$c(y, \phi) = -\log(y!)$$

Discrete distributions

2. Poisson distribution (continued)

$$\mathbb{E}(y) = b'(\eta) = \exp(\eta) = \lambda = \mu$$

$$\text{Var}(y) = \phi b''(\eta) = \phi \exp(\eta) = 1 \cdot \lambda = \lambda$$

Canonical link:

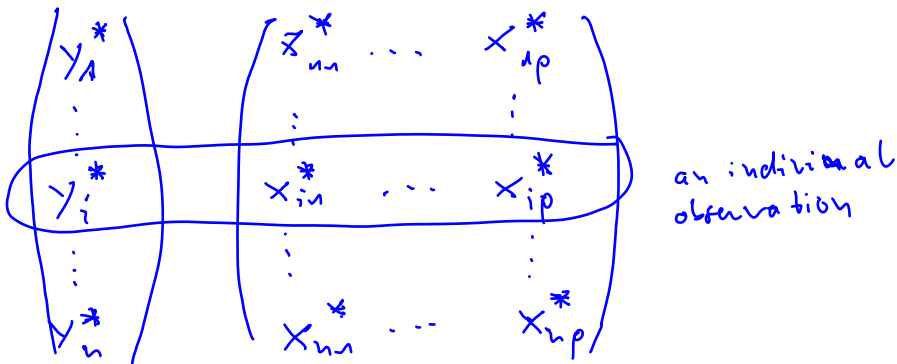
$$\eta = \log(\lambda) = \log(\mu) = \underline{x}^T \underline{\beta}$$

$$\Rightarrow \mu = \lambda = \exp(\underline{x}^T \underline{\beta})$$

Grouped data

notation: y_i^* individual data
 $\bar{y}_i$ certain group

- So far, we assumed individual data, i.e. one observation $(y_i^*, \mathbf{x}_i^*)$ is assigned to the respective object i :



Grouped data

- In the matrix of covariates there might be some identical rows. Then, the observations can be grouped as follows:

$$\begin{array}{l} \text{group 1} \\ \text{group } i \\ \text{group } G \end{array} \begin{pmatrix} n_1 \\ \vdots \\ n_i \\ \vdots \\ n_G \end{pmatrix} \begin{pmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_i \\ \vdots \\ \bar{y}_G \end{pmatrix} \begin{pmatrix} x_{11} & \dots & x_{1p} \\ \vdots & & \vdots \\ x_{i1} & \dots & x_{ip} \\ \vdots & & \vdots \\ x_{G1} & \dots & x_{Gp} \end{pmatrix}$$

- n_i : number of observations in group i
- $\bar{y}_i$: average response in group i
- $\mathbf{x}_i$: vector of covariates in group i

Grouped data

Binomial distribution

Example: Bernoulli distribution

$$y_1^*, \dots, y_n^* \in \{0, 1\} \quad y_i^* | \mathbf{x}_i^* \sim \text{Bernoulli}(\pi_i) = B(1, \pi_i)$$

$$\rightarrow \sum_{j=1}^{n_i} y_{ij}^* | \mathbf{x}_i \sim B(n_i, \pi_i) \quad \in \{0, \dots, n_i\}$$

$$\rightarrow \bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}^* \in \{0, \frac{1}{n_i}, \dots, 1\} \quad \text{scaled binomial dist.}$$

$$f_{\text{skal}}(\bar{y}_i) = P(\bar{Y}_i = \bar{y}_i) = \bar{y}_i \sim B(n_i, \pi_i) / n_i$$
$$f_{\text{Binom}}(n_i, \bar{y}_i)$$

$$= \binom{n_i}{n_i \bar{y}_i} \pi_i^{n_i \bar{y}_i} (1 - \pi_i)^{n_i - n_i \bar{y}_i}$$

$$= \exp \left(\frac{\bar{y}_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + \log (1 - \pi_i)}{1/n_i} + \log \binom{n_i}{n_i \bar{y}_i} \right)$$

Grouped data

Example: Bernoulli distribution (continued)

$$\eta_i = \log\left(\frac{\pi_i}{1-\pi_i}\right) \Leftrightarrow \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$b(\eta_i) = -\log(1-\pi_i) = \log(1 + \exp(\eta_i))$$



$$\phi_i = 1/n_i$$

$$c(\bar{y}_i, \phi_i) = \log\left(\binom{n_i}{n_i \bar{y}_i}\right)$$

Thus, we again have an exponential family for grouped data!
However, in this case the dispersion parameter ϕ_i depends on i .

Grouped data

General statement

For grouped data, still the exponential family is valid, but:

non-grouped y_i^*		grouped $\bar{y}_i$
ϑ_i	$\left. \begin{array}{l} \text{remain the same} \\ \text{after grouping it gets} \\ \text{scaled by } 1/n_i \end{array} \right\}$	ϑ_i
$b(\vartheta_i)$		$b(\vartheta_i)$
ϕ		ϕ/n_i
$c(y_i^*, \phi)$		$c(\bar{y}_i, \phi/n_i)$

have changed

Grouped data

In particular,

$$\mathbb{E}(\bar{y}_i) = b'(\eta_i) = \mathbb{E}[y_i^*] = \pi_i$$

$$\text{Var}(\bar{y}_i) = \frac{\phi}{n_i} b''(\eta_i) \Rightarrow \frac{1}{n_i} \text{Var}(y_i^*) \Rightarrow \frac{1}{n_i} \pi_i(1-\pi_i)$$

We use the notation $\phi_i = \frac{\phi}{n_i}$, i.e. ϕ_i depends on the observations just via n_i .

Grouped data

What actually is the advantage of grouped data?

- ▶ Completely equivalent w.r.t. estimation results
- ▶ But data partially is available as grouped data
- ▶ In some circumstances, grouping provides benefits:
 - ▶ better computation time and memory space
 - ▶ for model diagnosis



⇒ just the SEs of β_j 's change